Dong 2020 Prostate Cancer — analysis summary

RUN 3 OF 6

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

One connected kNN component; adeno (#1/#2/#5) intermix and are continuous with the NEPC pole (#6). iLISI rises 1.00 → 1.70 (ceiling 4) under Harmony θ=10, plateauing by θ=20; n_pcs=30 (robust band). Overturns the prior scalar-led NO-GO — the embedding is one continuum.

Gate 2 — module geography → GO (non-branching)

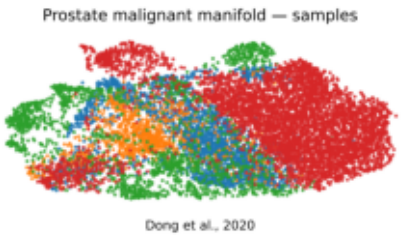
Single developmental axis: Gast & pSom co-vary (r=+0.91) at one pole; Crest inverse-gradient terminus. Crest opposes Gast (−0.34) and pSom (−0.48); joint pSom&Crest enrichment 0.66 (<1 → inverse, not a hub). No Gast arm exists to invert onto → no Crest-arm flag. Passes as the non-branching case.

Gate 3 — hub cross-repression → NA

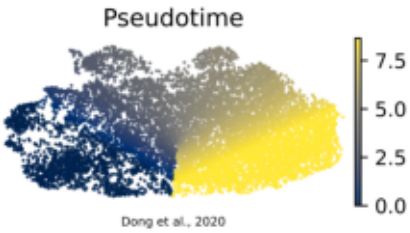
Committed principal curve is 2 tips / 0 forks. Branching check (shown, not asserted): a forced PPT tree (Nodes=200) gives 7 tips / 5 branch nodes — but node-density over-segmentation, not a bifurcation: branch count collapses 5→4→3→2→1 as PPT nodes drop 200→100→50→30→20, and the 7 tips co-vary in Gast & pSom (r=+0.90), not competing arms. No bifurcation → Gate 3 not applicable.

STEP FIGURES

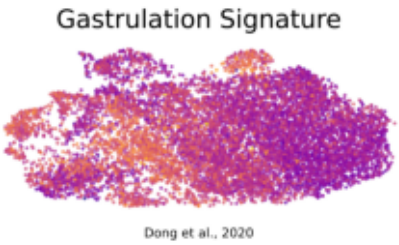
Manifold — samples



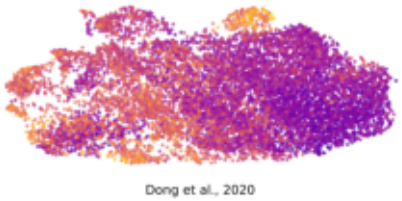
Pseudotime



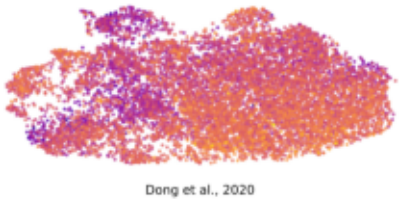
Gastrulation



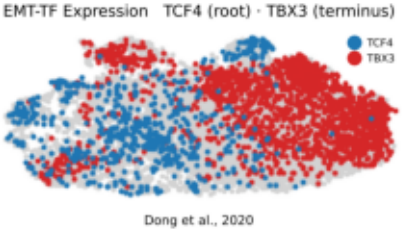
Somitogenesis



Neural Crest



EMT-TF expression



Dataset.
Primary prostate tumours (Dong et al., 2020), 10x — 5 adenocarcinoma + 1 small-cell NEPC; malignant compartment, 9,698 cells across 4 malignant-bearing samples.

Verdict.
The pattern reproduces as the NON-BRANCHING case — one gastrulation/somitogenesis-high origin (adeno) → neural-crest-high terminus (NEPC) axis, cross-repression collapsed. Geometry: Linear. Gate 1 GO / Gate 2 GO / Gate 3 NA. Module means Gast +0.035 / pSom +0.100 / Crest −0.041. EMT-TFs TCF4 (early/root) → TBX3 (late/terminus); Crest-high decile mean pseudotime 0.92 (terminal).

Parameter choices

Min-cell threshold	floor ≥100 (gap premise fails: counts 915–4765, gap at top)	**Samples kept**	4/4 malignant-bearing (of 6 total patients)
n_pcs / θ	30 (robust 20–40 band) / Harmony θ=10 (iLISI plateau)	**Geometry**	Linear — principal curve (ElPiGraph, 20 nodes, fold-free 2.64)
Root	Gast/pSom-high curve tip (developmental origin, Crest-low)	**Module means**	Gast +0.035 / pSom +0.100 / Crest −0.041 (raw score_genes)
EMT-TFs	TCF4 (early/root) → TBX3 (late/terminus), detection-gated		